

Tyler Walters

PORTLAND, OR - (503) 309-3201

resume@tylerwalts.com – (<https://www.linkedin.com/in/tylerwalts/>)

SUMMARY

Cloud Architect with over 20 years of expertise in designing, scoping, estimating, delivering and operating custom software and hardware solutions. A customer-obsessed technical generalist with demonstrated success in delivering high-impact results at companies ranging from seed-funded start-up to Fortune 50 leader. A continuous learner and thought leader with both deep expertise in industry foundations and the curiosity to experiment with leading-edge technologies. An approachable, hands-on technical leader who seeks diverse perspectives, generously supports and develops others, and understands the importance of being an emotionally-intelligent coach, peer, and mentor. Specialized in the following areas:

- Software Architecture
- Cloud Native Applications
- Custom App Development
- CI/CD Pipelines & DevOps
- Serverless & Microservices
- Large-Scale Services on AWS
- Security & Data Privacy
- Event-Driven Architecture
- Ops Metrics/Dashboards/Alarms
- Roadmap Planning
- Confident Communicator
- Public Speaker & Presenter
- Cross-Functional Collaborator
- Trainer & Mentor
- AWS-Certified (4)

PROFESSIONAL EXPERIENCE

Roar Social, Los Angeles, CA (Remote)

July 2022 - Present

Roar Social is a social network that gamifies giving by replacing “like” buttons with micro-donations that support charitable causes.

Software Architect

Responsible for defining App-to-API interactions, architecture design and leading implementation of backend services and a team of 25 software engineers. Transformed a POC demo app to a secure, scalable and observable production-ready app in 9 months. Iteratively delivered a multi-feed video discovery and streaming application, donation flow and administration system through the full lifecycle of defining scope, architecture design reviews, tasking, implementation, CI deployments, metric-driven operations and reporting. Load tested content delivery APIs up to 100K TPS at <50ms avg latency. Personally implemented critical components and reference examples, including:

- **Personalized Feed Generation:** Created an event-driven process to manage personal sets of content users have seen, viewed and interacted with on the system and ensure that blocked content is never seen again, duplicates are avoided and friend, follower and highly-scored content items are promoted. Planned a roadmap for A/B trial of a ML/AI-based approach.
- **Operational Excellence:** Led a program to ensure successful service operations using dashboards & metrics, runbooks, on-call shifts and an auto-generated ticketing system to quickly react to system metric anomalies and log errors.
- **Content Decoration:** Deployed a just-in-time metadata decorator for each content ID in a user's feed so that the stats were always fresh and personalized to show if/how a user interacted with a content item.
- **Admin UI:** Created internal tools to help visualize the state of each user's feed and how they see a decorated content item.
- **Content Discovery System:** Designed BFF and interaction between client and backend and implemented the APIs and data storage & caching systems.
- **High Performance:** Solved serverless performance issues by migrating key lambdas to a containerized NestJS service deployed to AWS Fargate. Also improved the developer experience by using containers for local development.
- **Daily Leadership:** Unblocked implementation escalations, PR Reviews, Pipeline health, Deployments, Jira Task creation & grooming, Eng task QA, Product planning & estimation, Recruiting & hiring Support

Amazon.com, Inc., Portland, OR

January 2018 - July 2022

Amazon.com, Inc. is a technology company focusing on e-commerce, cloud computing, advertising, streaming, and artificial intelligence.

Software Development Engineer: Selling Partner Services (<https://sellercentral.amazon.com>)

Managed 1.6 PB of 3rd-party Seller inventory data, continuously updated via events ingested from 14 high-volume sources and indexed into searchable and aggregated formats in a highly distributed and available data store.

- Led a campaign to modernize 160+ data clients across 90 teams through close coordination and automated status reporting.
- Scaled a data routing service to comfortably handle 340K TPS at 50ms latency using Go and ECS Fargate auto-scaling.
- Used metric data to drive service improvements that led to a 30% reduction in high-severity tickets.

Software Development Engineer: AWS Training & Certification (<https://www.aws.training>)

Responsible for the course catalog, survey and certification systems used to teach students and aspiring IT professionals how to use the AWS cloud in a variety of use case applications and become certified.

- Defined a roadmap for modernizing and scaling a C# monolith by replacing key workloads and APIs with microservices.
- Improved the Course Search UX, accuracy and performance by ingesting data into OpenSearch schemas.
- Scaled the LMS and dependent services prior to the annual re:Invent conference to handle up to 50K TPS.
- Owned department-level campaign to comply with EU cookie consent across all AWS-owned web properties in training.

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- Worked with 3rd parties Qualtrics and Certmetrics to reduce the number of downstream outages that impacted customers.

Cloud Application Architect: AWS Professional Services (<https://aws.amazon.com/professional-services>)

Provided consulting & professional services to help customers implement solutions powered by AWS. Grew the practice by giving interviews, mentoring new hires and creating delivery reference materials and tech presentations. Certified AWS Speaker. Led a mix of AWS and client staff to build and operate cloud-native applications for 3 to 9 month engagements:

- [Whole Foods Market eStore](#): Re-implemented the online catering ordering website, 'eStore' on AWS, compliant with internal Amazon best practices and integrated with internal Amazon systems after the Amazon acquisition of WF. Responsible for doing the initial assessment, overall design and leading the team to successfully implement an accelerated 9-month timeline.
- [DoorDash](#) Deployment Modernization: Replaced 3rd-party cloud deployment tooling and integrated Jenkins with AWS CodeDeploy on ECS for canary deployments, faster rollbacks and multi-region support.
- [Confidential Client] Drive-Through Automation R&D: Worked between AWS Lex and Data Science teams building a POC voice service for augmenting the drive-through experience. Responsible for building and operating a custom voice-UI integration testing framework and a pipeline that automated ingestion of audio utterances to create test data. The test platform was used as part of a feedback loop while training ML models.
- [Portland General Electric](#): Implemented AWS Landing Zone and Okta integration with IAM SSO to help bridge legacy datacenter with a multi-account cloud usage strategy.
- [Alexa Music Skills](#), Developer Experience Team: Helped deliver self-service music skill creation capability to Alexa, including support for the Apple Music provider launch, radio stations and music expansion into LatAm, EU and Asian marketplaces. Designed and implemented music skill metadata store and customer-specific music skill retrieval APIs.

Nike, Inc., Beaverton, OR

October 2016 - January 2018

Nike, Inc. is the world's largest supplier of athletic shoes and apparel headquartered near Beaverton, Oregon, United States.

Tech Lead, Trial Zone

The [Nike Trial Zone](#) is a set of digitally enhanced environments where customers can have AR experiences tailored to each type of shoe, watch videos of themselves and have selfie social posts. Increased customer sign-ups to the Nike+ loyalty program.

- Basketball: On a mini basketball court surrounded by hidden Kinect sensors, customers run signature moves that match the athlete the shoes were designed around (Jordans, Lebrons, etc) while following their progress on a TV wall behind the hoop.
- Running: On a treadmill facing a surrounding screen, customers run popular routes virtually while embedded cameras record the shoe performance. The videos from multiple angles are used to decide which shoe gives the best running experience.
- Joined Nike to take over the agency-created retail experience POC and create a team to secure, standardize and scale the system to 8 flagship Nike retail locations across US, Europe and China.
- Implemented CI/CD to maintain a fleet of on-site servers to support the tracking sensors, video streams and game engine. Used a set of cloud-based services to monitor performance and deliver new video content to go along with new shoe drops.
- Refactored Unity3D games to have configurable assets to support product and venue-specific experiences.
- Automated site-specific configuration to scale to more locations while expanding features and delivering fresh asset packs.

Slalom Consulting, Chicago, IL

April 2010 - October 2016

Slalom is a business and technology consulting firm headquartered in Seattle, Washington with annual revenue over two billion dollars.

Solution Principal

Managed a team of 9 SDE consultants, co-founded a DevOps focused practice and led client delivery of custom and cloud-native applications for enterprise clients:

- [Echo Global Logistics](#): Integrated an acquired stack of IT services using IaC approaches, Chef, .NET, Hyper-V, Azure, TerraForm and DataDog.
- [Western Governors University](#): Designed a Data CI pipeline for defining data as code using Liquibase (Datical), Chef, Vagrant and Jenkins.
- [Qlik Technologies](#): Technical lead to implement and initially operate QlikCloud.com until it reached 800 users, then transitioned ownership to Qlik-based team that was hired to take ownership. We took an enterprise hosted service and enabled it as an on-demand SaaS in less than a year. Delivered a globally distributed, HA and auto-scaling system based on concurrent user login count using MEAN stack for front-end; AWS, Puppet, Consul, New Relic, BlazeMeter & SauceLabs.
- [Pepsi](#): Enabled better business analysis of Pepsi vending machine metrics by defining Talend, RDS and Tableau servers as code using CloudFormation on AWS. Demonstrated how to ingest raw data, clean & transform it, then build dashboards.
- [Law Bulletin](#): Migrated JuraLaw legal docket management system from on-premise Oracle/Liferay/Solr/Jasper stack to AWS. Led the cloud vendor selection process and demonstrated better consistency, agility and sprint velocity using Puppet, CloudFormation & CI/CD. Implemented Infrastructure as Code (IaC) to ensure consistent environments and add DR ability.
- [Walgreens](#): Implemented a time-sensitive data de-identification system to meet compliance deadlines and avoid disruption of reporting services. Ensured that app data was scrubbed of identifying information before flowing between pharmacies.

- [Blue Cross Blue Shield of Michigan](#): Lead engineer on a new member portal for insurance members to manage accounts and claims. Scaled team to 22 developers by automating environment setup using RedHat/WebSphere/Oracle and an IaC & CI/CD approach to delivery. Used Backbone Js, Java Spring, Liquibase for MySQL/Oracle, CloudFormation & Puppet.
- [Providence Health & Services](#): Partnered with a new Director of Healthcare Intelligence to establish a system for applying BI principles, tooling and governance to operational and research data marts. Defined roles, processes and delivered the first two pilot projects to help analyze bundled knee surgery packages and stroke services using MSBI and Tableau.
- [StanCorp Financial Group, LLC](#): Worked with their BI team to introduce Microsoft BI stack (MSBI) and converted a set of underwriting reports migrated from Cognos. Automated server configuration as code and documented runbook processes. Trained the operating team and created a center of excellence to define security, governance and an end-user training plan.
- [Russell Investments](#): Created a Salesforce app in SFDC to help account managers to decide which client accounts to prioritize relationship management activities. Also created an overall dashboard view for management.
- [CorVel Software](#): Automated the transition from Sales to Engineering at CorVel by creating a form generation and workflow service to automate the on-boarding of each of their new customers using C# .NET.

[Hitachi Consulting](#), Portland, OR

August 2006 - April 2010

Hitachi Consulting was an international management and technology consulting firm founded by technology leaders from Grant Thornton.

Senior Consultant

IT consultant specializing in Application Development for enterprise clients. Project engagements included:

- [Waste Management](#): Upgraded, optimized, re-designed and integrated Greenopolis.com site with other WM web properties such as ThinkGreenRewards.WM.com. Added a portal interface to provide the UI for can and bottle recycling kiosks. Automated environment cloning of the Drupal LAMP stack using Drush and bash to ensure standard environments and DR.
- [Nike](#): Helped the Vendor Management office transition from a paper to digital process by customizing their existing SharePoint system. Used an IaC approach to defining a vendor management workflow system using SharePoint site templates, custom C# web parts and PowerShell scripts.
- [Microsoft](#): Prepared data visualization workshops and authored 2 Green IT white paper articles for VisioToolbox.com in preparation for a Green IT conference. Mocked out HVAC sensor data generation, indexed in MySQL and bound to Visio to visualize data center temperature heat maps used to make cooling more efficient.
- [Johnstone Supply](#): Enhanced data-driven marketing decisions and automated price setting for retail locations. Implemented MSBI stacks and configured daily intake, processing and analysis of retail store sales, costs and inventory CSVs using SSIS, SSAS and SSRS. Developed price setting UI to support region-specific campaigns based on predictive values.
- [Cascade Corporation](#): Helped Cascade comply with new EU regulations by customizing CNC tooling machine code to stamp load bearing data on forklift attachments to meet new regulations.
- [Columbia Sportswear](#): Updated corporate social responsibility app that enabled inspectors to remote factory locations offline, then automatically synchronize data when they come back online. Used C# .NET and XML-based data storage.
- [Knowledge Learning Corporation](#): Created kiosk time-clock software to allow parents and teachers to check children in/out of KinderCare locations. Also created an administration UI for compliance, performance and financial reporting using Java Struts and Oracle. Worked with Marketing and developed a set of reports to analyze marketing campaign effectiveness.

[WitMatix.com Inc.](#), Corvallis, OR

September 2004 - August 2006

WitMatix was a web development, e-commerce and hosting provider located in Corvallis, Oregon and a member of [AEP](#) at OSU.

Founder & Web Developer

Started a web development and online services company while in college and managed a team of local students and freelance contractors to create online presences and e-commerce systems for small business clients. Proposed, sold, designed, developed and managed project relations between my development group, 3rd-party designers, and clients. Client projects included:

- HeadTracker.com: Sold IR head tracking hardware to gamers and people with physical disabilities.
- SunRoseSheep.com: Informational website for a sheep farm and wool supplier.
- GetFlossin.com: Website, SEO, Advertisement landing page and contact form for coloring books for dental offices.
- BerkshireTownhomes.com: Website and contact form for apartment rentals.
- BuyGoodMusic.com: Online marketplace for independent music artists.
- TheChristianQuarter.com: Online dating service serving people who follow the Christian faith. Included a Java-based IM.
- RoachWear.biz Screen Printing: E-commerce store to sell custom printed shirts, hats and clothing.
- MottoRX.com Pharmacy: An e-commerce store selling products from a Canada-based pharmacy.
- RendezvousPCs.com: E-commerce store to sell PCs and related accessories.
- Supported other miscellaneous small website, hosting and domain name registration customers.

PEAK Internet, Corvallis, OR**June 2003 - September 2004***PEAK Internet delivers fixed wireless or fiber Internet with the option for optimized premium Wi-Fi to many underserved areas in Oregon.***Software Engineer**

Software development engineer for an Internet Services Provider (ISP) responsible for implementing online payments and a UX and backend for self-service billing and account management.

- Coordinated input from marketing, billing and IT departments to create initial mockups and the final implementation.
- Integrated with the 3rd party credit card payment system and handled customer data securely.
- Provided advanced support services for hosting and web customers using PHP, Perl, MySQL, Postgres, Apache & Linux.
- Built and racked physical servers and configured them in the ISP data center to connect to NTP, LDAP & other services.

Business Solutions Group, Corvallis, OR**March 2003 - June 2003***BSG was a college partnership program and a source for efficient, cost-effective outsourcing services to business clients.***Developer**

Worked in the Business Solutions Group (BSG) as part of a partnership between Oregon State University and Hewlett-Packard.

- Analyzed and tested network hardware and software for HP using custom C# & .NET applications.
- Performed load, fault and smoke testing and network traffic analysis on network switches and hubs..
- Tested software interfaces to hardware switches and hubs for usability and functionality.

Naval Air Warfare Center Training Systems Division (NAWCTSD), Orlando, FL **June 2002 - August 2002***The principal Navy center for research, development, test and evaluation, acquisition and product support of training systems.***Research IT Developer**

Advanced Distance Learning (ADL) is an initiative with a mission to standardize and distribute military training material.

- Worked with university research professors and a Navy liaison to help teachers create Advanced Distance Learning courses.
- Installed & configured Linux servers to host an XML-based course validation service built on the LAMP platform.
- Wrote an online survey using PHP to collect feedback from course creators and store it for querying and review.
- Created an online catalog to organize and browse ADL Guidelines to make it easier for teachers to consume than PDF.

Corvallis MicroTechnology (CMT), Corvallis, OR**September 2001 - June 2002***CMT is the leading manufacturer of hand-held computers for field professionals in forestry, utility, F&W and environmental assessment.***SysAdmin, QA Tester, Web Developer**

Managed IT systems and supported business operations for a surveyor-grade GPS company, Corvallis MicroTechnology.

- Maintained and improved the company website, FTP server, Perl Scripts, MySQL DBs and office network.
- Provided GPS hardware and software support to customers in person and remotely.
- Performed QA testing and troubleshooting on problematic GPS hardware.

Delta Systems Inc., Portland, OR**June 1998 - September 1998***Delta Systems was a business technology service provider to companies and government agencies in the Oregon area.***System Administrator**

Accelerated a plan to replace 400+ PC workstations for Oregon Public Employee Retirement Plan IT office building and finished 6 weeks ahead of schedule..

- Streamlined the process by creating two workstation assembly lines for setting up new PCs, and for processing old PCs.
- Configured new PCs to state government standards, connected them to the network and installed custom software according to the role of each user. Coordinated data migration and trained each user as needed.
- Refurbished existing PCs by wiping, installing operating systems and running test suites to verify they were ready for resale.

EDUCATION**Master of Business Administration (MBA), Oregon State University****Bachelor of Science, Computer Science, Oregon State University****CERTIFICATIONS****AWS Certified Solutions Architect****AWS Certified SysOps Administrator****AWS Certified Developer Associate****AWS Certified Cloud Practitioner****Microsoft MCPD: SharePoint Developer 2010****Microsoft MCTS: Microsoft SharePoint 2010, Application Development****Microsoft MCST: Microsoft Office SharePoint Server 2007 - Configuration**